YI-CHIEN CHIANG (江易謙)

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PROFESSIONAL SUMMARY

Dependable Software/Firmware Engineer with 2 years of experience in embedded systems and 1.5 years in Windows Presentation Foundation (WPF). Proficient in C/C++, C#, and Python. Hands-on experience deploying and configuring Ardupilot-based USVs, building Gazebo simulation platforms, applying reinforcement learning (TD3) for autonomous vehicle parameter tuning, and developing low-latency video streaming pipelines (GStreamer, RTP/RTSP). Demonstrated English proficiency (TOEFL iBT: 86) and a proven track record of delivering robust, high-performance solutions for aerospace and instrumentation applications.

WORK EXPERIENCE

Firmware Engineer

Taiwan Innovative Space Inc., Taipei, Taiwan — May 2023 – Present

- Developed and optimized firmware for rocket sensor systems, integrating accelerometers, strain gauges, and thermocouples to enable high-precision data acquisition and real-time processing for mission-critical aerospace applications.
- Ported C++ firmware to STM32 microcontrollers, enhancing peripheral integration (ADC, UART, SPI, I2C) and achieving modular, scalable codebases for multiple platform variants.
- Designed automated environmental stress screening (ESS) test software for rocket subsystems, streamlining validation under thermal, vibration, and vacuum conditions, improving testing efficiency by 50%.
- Collaborated with cross-functional teams to validate designs, analyze test results, and implement improvements, enhancing subsystem reliability.
- Developing the autopilot for an Ardupilot-based unmanned surface vehicle (USV): parameter tuning via a TD3 reinforcement-learning pipeline in Gazebo simulation, sensor calibration, and a custom Lua servo-failsafe extending Ardupilot's native behavior.

Software Engineer

Good Will Instrument Co., Ltd., Taipei, Taiwan — September 2021 – March 2023

- Developed user interface for digital storage oscilloscopes using WPF with MVVM architecture.
- Implemented protocol decoding and analysis for CAN Bus, LIN Bus, I2C, SPI and UART, enabling accurate signal visualization and debugging for end-users.
- Collaborated with hardware teams to integrate UI with oscilloscope firmware, ensuring seamless functionality across diverse device configurations.

Mechatronics Engineer

Huro Auto Co., Ltd., Taipei, Taiwan — March 2021 – September 2021

- Built and maintained mechatronics systems for electric buses, focusing on CAN Bus network integration to ensure reliable communication between control units.
- Conducted diagnostic testing and troubleshooting of CAN Bus systems.

PROJECTS

- **GStreamer UAV Video Link** (2026): Built a UAV video-link pipeline in C/GStreamer — camera → H.264/H.265 encode → RTP/RTSP → decode/display — with configurable encode presets, live FPS monitoring, and MAVLink VIDEO_STREAM_INFORMATION for QGroundControl auto-discovery; characterized software encoding CPU cost and latency across presets on ARM Cortex-A76 (H.264 ultrafast: ~10% CPU / 7 ms; H.265 ultrafast: ~100% CPU / 144 ms encode latency — motivating hardware encoder selection on Jetson/i.MX targets). [repo]
- **Gazebo USV Simulation Platform** (2025): Forked the open-source ASV wave simulation, enhanced hydrodynamic physics and runtime performance, and extended it into a full Gazebo-based USV simulation environment for hardware-free testing.
- **Pico C++ Firmware Template** (2025): Developed a general-purpose RP2040 firmware template integrating FreeRTOS, micro-ROS, and an MCP2515 CAN Bus driver with a task-based architecture for rapid embedded project setup.

- **Rocket Sensor Firmware** (2023): Developed C++ firmware for STM32-based rocket sensor system, optimizing real-time telemetry for aerospace applications.
- **Oscilloscope UI** (2022): Built WPF-based UI with MVVM for digital oscilloscope, improving signal visualization and user interaction.

EDUCATION

Bachelor of Science in Automation Engineering
National Formosa University, Yunlin County, Taiwan — September 2015 – June 2019

Vocational High School Diploma in Control Engineering
Taichung Industrial High School, Taichung City, Taiwan — September 2012 – July 2015

CERTIFICATIONS

- TOEFL iBT: Score of 86, Educational Testing Service (ETS), September 13, 2020

SKILLS

- **Programming:** C/C++, C#, Python
- **Frameworks:** WPF (MVVM), STM32, FreeRTOS, Ardupilot, ROS2, micro-ROS
- **Hardware:** RP2040 (Raspberry Pi Pico), MCP2515 CAN controller, Raspberry Pi 5 (ARM Cortex-A76)
- **Protocols:** CAN Bus, LIN Bus, I2C, SPI, UART, MAVLink
- **Video / Streaming:** GStreamer, RTP/RTSP, H.264/H.265 encoding
- **Simulation:** Gazebo, Ardupilot SITL, ASV Wave Simulation
- **ML / Robotics:** TD3 (Reinforcement Learning), Autonomous Vehicle Parameter Tuning
- **Tools:** Git, Visual Studio, STM32CubeIDE, CMake, Lua scripting
- **Soft Skills:** Problem-Solving, Team Collaboration